

11.1 Understanding the route to geographical enquiry and planning fieldwork

1 Put the following in order to show the steps to completing fieldwork:

- analyse results
- collect data
- decide what data to collect and how to collect it
- check results against hypothesis
- make conclusions
- form a hypothesis
- present results

2 Sort the following statements about setting an aim and formulating hypotheses into 'true' or 'false':

- the aim should be simple
- the aim should be complex
- the aim should identify many questions
- the aim should identify one question
- a hypothesis is a prediction
- hypotheses are always proven to be correct

3 Copy the following text and complete by choosing the correct word in the brackets.

Near the (**end/start**) of your fieldwork project, you should check if the data supports your hypothesis. The information you collect will prove, partially prove or (**support/disprove**) your hypothesis. If your data does not support your hypothesis, you start again, (**resetting/repeating**) a hypothesis and retesting.

When estimating, it is important to apply any scales, measurements or reference points the questions give you. Your answers to 'estimate' questions should be based on clear evidence and logical reasoning, rather than random guesses.

Questions that use the command word 'suggest' require you to apply your knowledge – remember that there is not one fixed answer, so you will have to use logical thinking.

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 A student has collected the data in Table 11.1 for a river study fieldwork:

Site	Depth of the river (cm)
A	5
B	6
C	7
D	12
E	15

Table 11.1: Data for a river study fieldwork

Estimate the average depth of the river. Choose the closest estimate from the list below:

a 4 cm

- b** 6 cm
- c** 9 cm
- d** 14 cm

[1]

5 A geography class is planning to investigate the number of people using a local park at different times of the day. Suggest a way to record and present this data. [4]

6 A student is carrying out research into waste collection services in different parts of a city in a low-income country (LIC). Their data can be found in Table 11.2.

Area	Waste collection service?
A	Yes
B	Yes
C	No
D	Yes
E	No

Table 11.2: Data about waste collection services in different parts of a city

Estimate the percentage of the city that has access to a waste collection service. Choose the closest estimate from the list below:

- a** 20%
- b** 40%
- c** 60%
- d** 80%

[1]

« RECALL AND CONNECT 1 «

Fieldwork can be carried out to test models and theories about rivers. One example is the Bradshaw model. See if you can answer these questions about the Bradshaw model linked to [Topic 1, 'Rivers'](#).

- 1 What is the Bradshaw model?
- 2 According to the Bradshaw model, how does discharge change from upstream to downstream?
- 3 According to the Bradshaw model, which river characteristics or features decrease from upstream to downstream?

UNDERSTAND THIS TERM

- Hypothesis

REFLECTION

You must plan and carry out fieldwork in a particular order – this is known as the route to geographical enquiry. Remembering the order is easier if you understand the process. For example, you cannot collect data until you have decided which methods of data collection to use. Creating a flow chart often helps to understand a process. Can you create a flow chart to show how you can plan and carry out fieldwork? How many stages would it have? What would each stage be?

- 7 Match the following terms, a–e, to their correct definition or example of the term, 1–5.

a	Hypothesis	1	Formed from a question that will be analysed using data.
b	Investigator bias	2	A statement that can be tested.
c	A statement that can be tested	3	A student only asking their friends in their class questions in a mobile phone survey.
d	Evaluation of a hypothesis	4	Choosing data that influences the outcome of a study.
e	Example of bias	5	Using collected data to determine if the hypothesis was true or false.

8 What makes a good hypothesis?

9 How does investigator bias affect research outcomes?

When devising a questionnaire, you need to ensure that your questions are relevant and clearly worded. This means avoiding unnecessary or vague questions and producing a questionnaire that is well-structured.

10 A student is investigating whether a local town has enough leisure facilities. Devise a questionnaire question that they could ask. [1]

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11 A student is investigating whether locals are happy with the number of tourists visiting their coastal town. Devise a questionnaire question that they could ask. [1]

12 A fieldwork group wants to test the hypothesis: 'People prefer to use public transport rather than private cars in city centres.' Devise a questionnaire question that they could ask to test this hypothesis. [1]

REFLECTION

A good hypothesis is simple and testable. Can you think of a hypothesis that can be created about coastal erosion? How would you devise an overall aim or question for an investigation about coastal erosion, and how would you then turn this into a hypothesis?